

One-Dimensional Maps

THE FUNCTION $f(x) = 2x$ is a rule that assigns to each number x a number twice as large. This is a simple mathematical model. We might imagine that x denotes the population of bacteria in a laboratory culture and that $f(x)$ denotes the population one hour later. Then the rule expresses the fact that the population doubles every hour. If the culture has an initial population of 10,000 bacteria, then after one hour there will be $f(10,000) = 20,000$ bacteria, after two hours there will be $f(f(10,000)) = 40,000$ bacteria, and so on.

A **dynamical system** consists of a set of possible states, together with a rule that determines the present state in terms of past states. In the previous paragraph, we discussed a simple dynamical system whose states are population levels, that change with time under the rule $x_n = f(x_{n-1}) = 2x_{n-1}$. Here the variable n stands for time, and x_n designates the population at time n . We will require that the rule be deterministic, which means that we can determine the present state (population, for example) uniquely from the past states.

No randomness is allowed in our definition of a deterministic dynamical system. A possible mathematical model for the price of gold as a function of time would be to predict today's price to be yesterday's price plus or minus one dollar, with the two possibilities equally likely. Instead of a dynamical system, this model would be called a random, or stochastic, process. A typical realization of such a model could be achieved by flipping a fair coin each day to determine the new price. This type of model is not deterministic, and is ruled out by our definition of dynamical system.

We will emphasize two types of dynamical systems. If the rule is applied at discrete times, it is called a discrete-time dynamical system. A discrete-time system takes the current state as input and updates the situation by producing a new state as output. By the state of the system, we mean whatever information is needed so that the rule may be applied. In the first example above, the state is the population size. The rule replaces the current population with the population one hour later. We will spend most of Chapter 1 examining discrete-time systems, also called maps.

The other important type of dynamical system is essentially the limit of discrete systems with smaller and smaller updating times. The governing rule in that case becomes a set of differential equations, and the term continuous-time dynamical system is sometimes used. Many of the phenomena we want to explain are easier to describe and understand in the context of maps; however, since the time of Newton the scientific view has been that nature has arranged itself to be most easily modeled by differential equations. After studying discrete systems thoroughly, we will turn to continuous systems in Chapter 7.

1.1 ONE-D I M E N S I O N A L M A P S

One of the goals of science is to predict how a system will evolve as time progresses. In our first example, the population evolves by a single rule. The output of the rule is used as the input value for the next hour, and the same

rule of doubling is applied again. The evolution of this dynamical process is reflected by composition of the function f . Define $f^2(x) = f(f(x))$ and in general, define $f^k(x)$ to be the result of applying the function f to the initial state k times. Given an initial value of x , we want to know about $f^k(x)$ for large k . For the above example, it is clear that if the initial value of x is greater than zero, the population will grow without bound. This type of expansion, in which the population is multiplied by a constant factor per unit of time, is called exponential growth. The factor in this example is 2.

WHY STUDY MODELS?

We study models because they suggest how real-world processes behave. In this chapter we study extremely simple models. Every model of a physical process is at best an idealization. The goal of a model is to capture some feature of the physical process. The feature we want to capture now is the patterns of points on an orbit. In particular, we will find that the patterns are sometimes simple, and sometimes quite complicated, or “chaotic”, even for simple maps. The question to ask about a model is whether the behavior it exhibits is because of its simplifications or if it captures the behavior despite the simplifications. Modeling reality too closely may result in an intractable model about which little can be learned. Model building is an art. Here we try to get a handle on possible behaviors of maps by considering the simplest ones. The fact that real habitats have finite resources lies in opposition to the concept of exponential population increase. From the time of Malthus (Malthus, 1798), the fact that there are limits to growth has been well appreciated. Population growth corresponding to multiplication by a constant factor cannot continue forever. At some point the resources of the environment will become compromised by the increased population, and the growth will slow to something less than exponential. In other words, although the rule $f(x) = 2x$ may be correct for a certain range of populations, it may lose its applicability in other ranges. An improved model, to be used for a resource-limited population, might be given by $g(x) = 2x(1 - x)$, where x is measured in millions. In this model, the initial population of 10,000 corresponds to $x = 0.01$ million. When the population x is small, the factor $(1 - x)$ is close to one, and $g(x)$ closely resembles the doubling function $f(x)$. On the other hand, if the population x is far from zero, then $g(x)$ is no longer proportional to the population x but to the product of x and the “remaining space” $(1 - x)$. This is

a nonlinear effect, and the model given by $g(x)$ is an example of a logistic growth model. Using a calculator, investigate the difference in outcomes imposed by the models $f(x)$ and $g(x)$. Start with a small value, say $x = 0.01$, and compute $f^k(x)$ and $g^k(x)$ for successive values of k . The results for the models are shown in Table 1.1. One can see that for $g(x)$, there is computational evidence that the population approaches an eventual limiting size, which we would call a steady-state population for the model $g(x)$. Later in this section, using some elementary calculus, we’ll see how to verify this conjecture (Theorem 1.5). There are obvious differences between the behavior of the population size under the two models, $f(x)$ and $g(x)$. Under the dynamical system $f(x)$, the

starting population size $x = 0.01$ results in arbitrarily large populations as time progresses. Under the system $g(x)$, the same starting size $x = 0.01$ progresses in a strikingly similar way at first, approximately doubling each hour. Eventually, however, a limiting size is reached. In this case, the population saturates at $x = 0.50$ (one-half million), and then never changes again. So one great improvement of the logistic model $g(x)$ is that populations can have a finite limit. But there is a second improvement contained in $g(x)$. If

we use starting populations other than $x = 0.01$, the same limiting population $x = 0.50$ will be achieved. **C O M P U T E R E X P E R I M E N T 1 . 1** Confirm the fact that populations evolving under the rule $g(x) = 2x(1-x)$ prefer to reach the population 0.5. Use a calculator or computer program, and try starting populations x_0 between 0.0 and 1.0. Calculate $x_1 = g(x_0)$, $x_2 = g(x_1)$, etc. and allow the population to reach a limiting size. You will find that the size $x = 0.50$ eventually “attracts” any of these starting populations. Our numerical experiment suggests that this population model has a natural built-in carrying capacity. This property corresponds to one of the many ways that scientists believe populations should behave—that they reach a steady-state which is somehow compatible with the available environmental resources. The limiting population $x = 0.50$ for the logistic model is an example of a fixed point of a discrete-time dynamical system. **Definition 1.1** A function whose domain (input) space and range (output) space are the same will be called a map. Let x be a point and let f be a map. The orbit of x under f is the set of points $x, f(x), f^2(x), \dots$. The starting point x for the orbit is called the initial value of the orbit. A point p is a fixed point of the map f if $f(p) = p$. For example, the function $g(x) = 2x(1-x)$ from the real line to itself is a map. The orbit of $x = 0.01$ under g is $0.01, 0.0198, 0.0388, \dots$, and the fixed points of g are $x = 0$ and $x = 1$. **2. 1. 2 COBWEB P L O T: G R A P H I C A L R E P R E S E N T A T I O N O F A N O R B I T** For a map of the real line, a rough plot of an orbit—called a cobweb plot—can be made using the following graphical technique. Sketch the graph of the function f together with the diagonal line $y = x$. In Figure 1.1, the example $f(x) = 2x$ and the diagonal are sketched. The first thing that is clear from such a picture is the location of fixed points of f . At any intersection of $y = f(x)$ with the line $y = x$,

the input value x and the output $f(x)$ are identical, so such an x is a fixed point. Figure 1.1 shows that the only fixed point of $f(x) = 2x$ is $x = 0$. Sketching the orbit of a given initial condition is done as follows. Starting with the input value $x = .01$, the output $f(.01)$ is found by plotting the value of the function above .01. In Figure 1.1, the output value is .02. Next, to find $f(.02)$, it is necessary to consider .02 as the new input value. In order to turn an output value into an input value, draw a horizontal line from the input–output pair $(.01, .02)$ to the diagonal line $y = x$. In Figure 1.1, there is a vertical dotted line segment starting at $x = .01$, representing the function evaluation, and then a horizontal dotted segment which effectively turns the output into an input so that the process can be repeated. Then start over with the new value $x = .02$, and draw a new pair of vertical and horizontal dotted segments. We find $f(f(.01)) = f(.02) = .04$ on the graph of f , and move horizontally to move output to the input

position. Continuing in this way, a graphical history of the orbit $.01, .02, .04, . . .$ is constructed by the path of dotted line segments. **EXAMPLE 1.2** A more interesting example is the map $g(x) = 2x(1-x)$. First we find fixed points by solving the equation $x = 2x(1-x)$. There are two solutions, $x = 0$

COBWEB PLOT A cobweb plot illustrates convergence to an attracting fixed point of $g(x) = 2x(1-x)$. Let $x_0 = 0.1$ be the initial condition. Then the first iterate is $x_1 = g(x_0) = 0.18$. Note that the point (x_0, x_1) lies on the function graph, and (x_1, x_1) lies on the diagonal line. Connect these points with a horizontal dotted line to make a path. Then find $x_2 = g(x_1) = 0.2952$, and continue the path with a vertical dotted line to (x_1, x_2) and with a horizontal dotted line to (x_2, x_2) . An entire orbit can be mapped out this way. In this case it is clear from the geometry that the orbit we are following will converge to the intersection of the curve and the diagonal, $x = 1/2$. What happens if instead we start with $x_0 = 0.8$? These are examples of simple cobweb plots. They can be much more complicated, as we shall see later.

and $x = 1/2$, which are the two fixed points of g . Contrast this with a linear map which, except for the case of the identity $f(x) = x$, has only one fixed point $x = 0$. What is the behavior of orbits of g ? The graphical representation of the orbit with initial value $x = 0.1$ is drawn in Figure 1.2. It is clear from the figure that the orbit, instead of diverging to infinity as in Figure 1.1, is converging to the fixed point $x = 1/2$. Thus the orbit with initial condition $x = 0.1$ gets stuck, and cannot move beyond the fixed point $x = 0.5$. A simple rule of thumb for following the graphical representation of an orbit: If the graph is above the diagonal line $y = x$, the orbit will move to the right; if the graph is below the line, the orbit moves to the left. **EXAMPLE 1.3** Let f be the map of given by $f(x) = (3x - x^3)/2$. Figure 1.3 shows a graphical representations of two orbits, with initial values $x = 1.6$ and 1.8 , respectively. The former orbit appears to converge to the fixed point $x = 1$ as the map is iterated; the latter converges to the fixed point $x = 1$.

Fixed points are found by solving the equation $f(x) = x$. The map has three fixed points, namely $1, 0$, and -1 . However, orbits beginning near, but not precisely on, each of the fixed points act differently. You may be able to convince yourself, using the graphical representation technique, that initial values near 1 stay near 1 upon iteration by the map, and that initial values near -1 stay near -1 . On the other hand, initial values near 0 depart from the area near 0 . For example, to four significant digits, $f(1) = 0.1495$, $f_2(1) = 0.2226$, $f_5(1) = 0.6587$, and so on. The problem with points near 0 is that f magnifies them by a factor larger than one. For example, the point $x = .1$ is moved by f to approximately $.1495$, a magnification factor of 1.495 . This magnification factor turns out to be approximately the derivative $f'(0) = 1.5$. **1.3 STABILITY OF FIXED POINTS** With the geometric intuition gained from Figures 1.1, 1.2, and 1.3, we can describe the idea of stability of fixed points. Assuming that the discrete-time system exists to model real phenomena, not all fixed points are alike. A stable fixed point has the property that points near it are moved even closer to the fixed point under the dynamical system. For an unstable fixed point, nearby points move away as time progresses. A good analogy is that a ball at the bottom of a valley is stable, while a ball balanced at the tip of a mountain

is unstable. The question of stability is significant because a real-world system is constantly subject to small perturbations. Therefore a steady state observed in a realistic system must correspond to a stable fixed point. If the fixed point is unstable, small errors or perturbations in the state would cause the orbit to move away from the fixed point, which would then not be observed. Example 1.3 gave some insight into the question of stability. The derivative of the map at a fixed point p is a measure of how the distance between p and a nearby point is magnified or shrunk by f . That is, the points 0 and .1 begin exactly .1 units apart. After applying the rule f to both points, the distance separating the points is changed by a factor of approximately $f'(0)$. We want to call the fixed point 0 “unstable” when points very near 0 tend to move away from 0. The concept of “near” is made precise by referring to all real numbers within a distance ϵ of p as the epsilon neighborhood $N_\epsilon(p)$. Denote the real line by $\mathbb{R}$. Then $N_\epsilon(p)$ is the interval of numbers $x : |x - p| < \epsilon$. We usually think of ϵ as a small, positive number. Definition 1.4 Let f be a map on $\mathbb{R}$ and let p be a real number such that $f(p) = p$. If all points sufficiently close to p are attracted to p , then p is called a

sink or an attracting fixed point. More precisely, if there is an $\epsilon > 0$ such that for all x in the epsilon neighborhood $N_\epsilon(p)$, $\lim_{k \rightarrow \infty} f^k(x) = p$, then p is a sink. If all points sufficiently close to p are repelled from p , then p is called a source or a repelling fixed point. More precisely, if there is an epsilon neighborhood $N_\epsilon(p)$ such that each x in $N_\epsilon(p)$ except for p itself eventually maps outside of $N_\epsilon(p)$, then p is a source. In this text, unless otherwise stated, we will deal with functions for which derivatives of all orders exist and are continuous functions. We will call this type of function a smooth function. Theorem 1.5 Let f be a (smooth) map on $\mathbb{R}$, and assume that p is a fixed point of f . 1. If $|f'(p)| < 1$, then p is a sink. 2. If $|f'(p)| > 1$, then p is a source. Proof: PART 1. Let a be any number between $|f'(p)|$ and 1; for example, a could be chosen to be $(1 + |f'(p)|)/2$. Since $\lim_{x \rightarrow p} \frac{f(x) - f(p)}{x - p} = f'(p)$, there is a neighborhood $N_\epsilon(p)$ for some $\epsilon > 0$ so that $|f(x) - f(p)| < a|x - p|$ for x in $N_\epsilon(p)$, $x \neq p$. In other words, $f(x)$ is closer to p than x is, by at least a factor of a (which is less than 1). This implies two things: First, if $x \in N_\epsilon(p)$, then $f(x) \in N_\epsilon(p)$; that means that if x is within ϵ of p , then so is $f(x)$, and by repeating the argument, so are $f^2(x)$, $f^3(x)$, and so forth. Second, it follows that $|f^k(x) - p| < a^k|x - p|$ (1.1) for all $k \geq 1$. Thus p is a sink. EXERCISE T1.1 Show that inequality (1.1) holds for $k \geq 2$. Then carry out the mathematical induction argument to show that it holds for all $k = 1, 2, 3, \dots$

EXERCISE T1.2 Use the ideas of the proof of Part 1 of Theorem 1.5 to prove Part 2. Note that the proof of part 1 of Theorem 1.5 says something about the rate of convergence of $f^k(x)$ to p . The fact that $|f^k(x) - p| < a^k|x - p|$ for $a < 1$ is described by saying that $f^k(x)$ converges exponentially to p as $k \rightarrow \infty$. Our definition of a fixed-point sink requires only that the sink attract some epsilon neighborhood $(p - \epsilon, p + \epsilon)$ of nearby points. As far as the definition is concerned, the radius ϵ of the neighborhood, although nonzero, could be extremely small. On the other hand, sinks often attract orbits from a large set of nearby initial conditions. We will refer to the set of initial conditions whose orbits converge to

the sink as the basin of the sink. With Theorem 1.5 and our new terminology, we can return to Example 1.2, an example of a logistic model. Setting $x = g(x) = 2x(1-x)$ shows that the fixed points are 0 and $1/2$. Taking derivatives, we get $g'(0) = 2$ and $g'(1/2) = 0$. Theorem 1.5 shows that $x = 1/2$ is a sink, which confirms our suspicions from Table 1.1. On the other hand, $x = 0$ is a source. Points near 0 are repelled from 0 upon application of g . In fact, points near 0 are repelled at an exponential magnification factor of approximately 2 (check this number with a calculator). These points are attracted to the sink $x = 1/2$. What is the basin of the sink $x = 1/2$ in Example 1.2? The point 0 does not belong, since it is a fixed point. Also, 1 does not belong, since $g(1) = 0$ and further iterations cannot budge it. However, all initial conditions from the interval $(0, 1)$ will produce orbits that converge to the sink. You should sketch a graph of $g(x)$ as in Figure 1.1 and use the idea of the cobweb plot to convince yourself of this fact. There is a second way to show that the basin of $x = 1/2$ is $(0, 1)$, which is quicker and trickier but far less general. That is to use algebra (not geometry) to compare $|g(x) - 1/2|$ to $|x - 1/2|$. If the former is smaller than the latter, it means the orbit is getting closer to $1/2$. The algebra says: $|g(x) - 1/2| = |2x(1-x) - 1/2| = |2x - 2x^2 - 1/2| = |2x - 1/2 - 2x^2| = |2x - 1/2| \cdot |1 - 2x|$. (1.2) Now we can see that if $x \in (0, 1)$, the multiplier $|1 - 2x|$ is smaller than one. Any point x in $(0, 1)$ will have its distance from $x = 1/2$ decreased on each iteration by g . Notice that the algebra also tells us what happens for initial conditions outside of $(0, 1)$: they will never converge to the sink $x = 1/2$. Therefore the

basin of the sink is exactly the open interval $(0, 1)$. Informally, we could also say that the basin of infinity is $(-\infty, 0) \cup (1, \infty)$, since the orbit of each initial condition in this set tends toward $-\infty$. Theorem 1.5 also clarifies Example 1.3, which is the map $f(x) = (3x - x^3)/2$. The fixed points are $-1, 0$, and 1 , and the derivatives are $f'(-1) = 0$, and $f'(0) = 1.5$. By the theorem, the fixed points -1 and 1 are attracting fixed points, and 0 is a repelling fixed point. Let's try to determine the basins of the two sinks. Example 1.3 is already significantly more complicated than Example 1.2, and we will have to be satisfied with an incomplete answer. We will consider the sink $x = 1$; the other sink has very similar properties by the symmetry of the situation. First, cobweb plots (see Figure 1.3) convince us that the interval $I_1 = (0, 3)$ of initial conditions belongs to the basin of $x = 1$. (Note that $f(-3) = f(3) = 0$.) So far it is similar to the previous example. Have we found the entire basin? Not quite. Initial conditions from the interval $I_2 = [2, 3)$ map to $(0, 1]$, which we already know are basin points. (Note that $f(2) = 1$.) Since points that map to basin points are basin points as well, we know that the set $[2, 3) \cup (0, 3)$ is included in the basin of $x = 1$. Now you may be willing to admit that the basin can be quite a complicated creature, because the graph shows that there is a small interval I_3 of points to the right of $x = 2$ that map into the interval $I_2 = [2, 3)$, and are therefore in the basin, then a small interval I_4 to the left of $x = 2$ that maps into I_3 , and so forth ad infinitum. These intervals are all separate (they don't overlap), and the gaps between them consist of similar intervals belonging to the basin of the other sink $x = -1$. The intervals I_n get smaller with increasing n , and all of them lie between -5 and 5 . Since $f(5) = -5$ and $f(-5) = 5$, neither of these numbers is in the basin of either sink.

EXERCISE T1.3 Solve the inequality $|f(x) - 0| > |x - 0|$, where $f(x) = (3x - 2)x$. This identifies points whose distance from 0 increases on each iteration. Use the result to find a large set of initial conditions that do not converge to any sink of f . There is one case that is not covered by Theorem 1.5. The stability of a fixed point p cannot be determined solely by the derivative when $|f'(p)| = 1$ (see Exercise 1.2).

So far we have seen the important role of fixed points in determining the behavior of orbits of maps. If the fixed point is a sink, it provides the final state for the orbits of many nearby initial conditions. For the linear map $f(x) = ax$ with $|a| < 1$, the sink $x = 0$ attracts all initial conditions. In Examples 1.2 and 1.3, the sinks attract large sets of initial conditions. EXERCISE T1.4 Let p be a fixed point of a map f . Given some $\epsilon > 0$, find a geometric condition under which all points x in $N_\epsilon(p)$ are in the basin of p . Use cobweb plot analysis to explain your reasoning. 1.4 PERIODIC POINTS Changing a , the constant of proportionality in the logistic map $g_a(x) = ax(1-x)$, can result in a picture quite different from Example 1.2. When $a = 3.3$, the fixed points are $x = 0$ and $x = 23/33 \approx .696969 \dots$, both of which are repellers. Now that there are no fixed points around that can attract orbits, where do they go? Use a calculator to convince yourself that for almost every choice of initial condition, the orbit settles into a pattern of alternating values $p_1 \approx .4794$ and $p_2 \approx .8236$ (to four decimal place accuracy). Some typical orbits are shown in Table 1.2. The orbit with initial condition 0.2 is graphed in Figure 1.4. This figure shows typical behavior of an orbit converging to a period-2 sink p_1, p_2 . It is attracted to p_1 every two iterates, and to p_2 on alternate iterates. There are actually two important parts of this fact. First, there is the apparent coincidence that $g(p_1) = p_2$ and $g(p_2) = p_1$. Another way to look at this is that $g^2(p_1) = p_1$; thus p_1 is a fixed point of $h = g^2$. (The same could be said for p_2 .) Second, this periodic oscillation between p_1 and p_2 is stable, and attracts orbits. This fact means that periodic behavior will show up in a physical system modeled by g . The pair p_1, p_2 is an example of a periodic orbit. Definition 1.6 Let f be a map on $\mathbb{R}$. We call p a periodic point of period k if $f^k(p) = p$, and if k is the smallest such positive integer. The orbit with initial point p (which consists of k points) is called a periodic orbit of period k . We will often use the abbreviated terms period- k point and period- k orbit. Notice that we have defined the period of an orbit to be the minimum number of iterates required for the orbit to repeat a point. If p is a periodic point of period 2 for the map f , then p is a fixed point of the map $h = f^2$. However, the converse is not true. A fixed point of $h = f^2$ may also be a fixed point of a lower

iterate of f , specifically f , and so may not be a periodic point of period two. For example, if p is a fixed point of f , it will be a fixed point of f^2 but not, according to our definition, a period-two point of f . EXAMPLE 1.7 Consider the map defined by $f(x) = x$ on $\mathbb{R}$. This map has one fixed point, at $x = 0$. Every other real number is a period-two point, because f^2 is the identity map. EXERCISE T1.5 The map $f(x) = 2x^2 - 5x$ on $\mathbb{R}$ has fixed points at $x = 0$ and $x = 3$. Find a period-two orbit for f by solving $f^2(x) = x$ for x . What about the stability of periodic orbits? As in the fixed point case, points near the periodic orbit can

be trapped or repelled by the orbit. The key fact is that a periodic point for f is a fixed point for f^k . We can use Theorem 1.5 to investigate the stability of a periodic orbit. For a period- k orbit, we apply Theorem 1.5 to the map f^k instead of f .

Definition 1.8 Let f be a map and assume that p is a period- k point. The period- k orbit of p is a periodic sink if p is a sink for the map f^k . The orbit of p is a periodic source if p is a source for the map f^k . It is helpful to review the chain rule of calculus, which shows how to expand the derivative of a composition of functions: $(f \circ g)'(x) = f'(g(x))g'(x)$ (1.3). Our current interest in the chain rule is for f^2 , in which case we have $(f^2)'(x) = f'(f(x))f'(x)$. If x happens to be a period-two point for f , the chain rule is saying something quite simple: the derivative of f^2 at a point of a period-two orbit is simply the product of the derivatives of f at the two points in the orbit. In particular, the derivative of f^2 is the same, when evaluated at either point of the orbit. This agreement means that it makes sense to talk about the stability of a period-two orbit. Now the period-two behavior of $g(x) = 3.5x(1-x)$ we found in Table 1.2 can be completely explained. The periodic orbit $.4794, .8236$ will be a sink as long as the derivative $(g^2)'(p_1) = g'(p_1)g'(p_2) = (g^2)'(p_2)$ is smaller than 1 in absolute value. A neasy calculation shows this number to be $g'(.4794)g'(.8236) = 0.2904$.

If instead we consider yet another version of the logistic map, $g(x) = 3.5x(1-x)$, the situation is again changed. The fixed points are $x = 0$ and $x = 5/7$. Checking derivatives, $g'(0) = 3.5$ and $g'(5/7) = 1.5$, so they are sources. The orbit $3/7, 6/7$ is a period-two orbit for g . Check that $(g^2)'$ at each of the orbit points is $5/4$, so that this period-two orbit repels nearby points. Now where do points end up?

C O M P U T E R E X P E R I M E N T 1 . 2 Write a computer program with the goal of redoing Table 1.2 for the logistic map $g_a(x) = ax(1-x)$, using $a = 3.5$. What periodic behavior wins out in the long run? Try several different initial conditions to explore the basin of the attracting periodic behavior. Then try different values of $a = 3.57$ and report your results. Now that we have some intuition from period-two orbits, we note that the situation is essentially the same for higher periods. Let $p_1, \dots, p_k$ denote a period- k orbit of f . The chain rule says that $(f^k)'(p_1) = (f'(f^{k-1}(p_1))) \dots f'(f(p_1)) = f'(f^{k-1}(p_1)) \dots f'(f(p_1))f'(p_1) = (f^k)'(p_k)$. (1.4)

STABILITY TEST FOR PERIODIC ORBITS The periodic orbit $p_1, \dots, p_k$ is a sink if $|(f^k)'(p_1)| < 1$ and a source if $|(f^k)'(p_1)| > 1$.

This formula tells us that the derivative of the k th iterate f^k of f at a point of a period- k orbit is the product of the derivatives of f at the k points of the orbit. In particular, stability is a collective property of the periodic orbit, in that $(f^k)'(p_i) = (f^k)'(p_j)$ for all i and j .

1 . 5 THE FAMILY OF LOGISTIC MAPS We are beginning to get an overall view of the family $g_a(x) = ax(1-x)$ associated with the logistic model. When $0 < a < 1$, the map has a sink at $x = 0$, and we will see later that every initial condition between 0 and 1 is attracted to this sink. (In other words, with small reproduction rates, small populations tend to die out.) The graph of the map is shown in Figure 1.5(a). If $1 < a < 3$, the map, shown in Figure 1.5(b), has a sink at $x = (a-1)/a$, since the magnitude of the derivative is less than 1. (Small populations grow to a steady state of x

(a - 1) a.) For a greater than 3, as in Figure 1.5(c), the fixed point $x = (a - 1) a$ is unstable since $g'(x) = a - 1$, and a period-two sink takes its place, which we saw in Table 1.2 for $a = 3.3$. When a grows above $1 + 6 = 3.45$, the period-two sink also becomes unstable. EXERCISE T1.6 Verify the statements in the previous paragraph by solving for the fixed points and period-two points of $g_a(x)$ and evaluating their stability.

C O M P U T E R E X P E R I M E N T 1 . 3 Use your logistic map program to investigate the long-run behavior of g_a for a near $a = 1 + 6$. Repeat Table 1.2 for values of a slightly smaller than a . What qualitative or quantitative conclusions can be made about the speed of convergence to the period-two orbit as a gets closer to a ? What happens to iterations beginning at a period-two point for a slightly larger than a ? For slightly larger values of a , the story of the periodic points of $g_a(x)$ becomes significantly more complicated. Many new periodic orbits come into existence as a is increased from 3.45 to 4. Figure 1.6 shows the limiting behavior of orbits for values of a in the range $1 + 6 \leq a \leq 4$. This computer-generated picture was made by repeating the following procedure: (1) Choose a value of a , starting with $a = 1 + 6$, (2) Choose x at random in $[0, 1]$, (3) Calculate the orbit of x under $g_a(x)$, (4) Ignore the first 100 iterates and plot the orbit beginning with iterate 101. Then increment a and begin the procedure again. The points that are plotted will (within the resolution of the picture) approximate either fixed or periodic sinks or other attracting sets. This figure is called a bifurcation diagram and shows the birth, evolution, and death of attracting sets. The term “bifurcation” refers to significant changes in the set of fixed or periodic points or other sets of dynamic interest. We will study bifurcations in detail in Chapter 11. We see, for example, that the vertical slice $a = 3.4$ of Figure 1.6 intersects the diagram in the two points of a period-two sink. For a slightly larger than 3.45, there appears to be a period-four sink. In fact, there is an entire sequence of periodic sinks, one for each period 2^n , $n = 1, 2, 3, \dots$. Such a sequence is called a “period-doubling cascade”. The phenomenon of cascades is the subject of Chapter 12. Figure 1.7 shows portions of the bifurcation diagram in detail. Magnification near a period-three sink, in Figure 1.7(b) hints at further period-doublings that are invisible in Figure 1.6. For other values of the parameter a , the orbit appears to randomly fill out the entire interval $[0, 1]$, or a subinterval. A typical cobweb plot formed for $a = 3.86$ is shown in Figure 1.8. These attracting sets, called “chaotic attractors”, are harder to describe than periodic sinks. We will try to unlock some of their secrets in later chapters. As we shall see, it is a characteristic of chaotic attractors that they can abruptly appear or disappear, or change size discontinuously. This phenomenon, called a “crisis”, is apparent at various a values. In particular, at $a = 4$, there is a

crisis at which the chaotic attractor disappears. For $a = 4$, there is no attracting set. The successive blow-ups of the bifurcation diagrams reveal another interesting feature, that of “periodic windows”. The period-three window, for example, is apparent in Figure 1.7(a) and is shown in magnified form in Figure 1.7(b). This refers to a set of parameter values for which there is a periodic sink, in this case a period-three sink. Since a period-three point of g_a is a fixed

point of the third iterate $g^3 a$, the creation of the period-three sink can be seen by viewing the de-

velopment of the graph of $g^3 a$ as a moves from 3.82 to 3.86. This development is shown in Figure 1.9. In Figure 1.9(a), the period-three orbit does not exist. This parameter value $a = 3.82$ corresponds to the left end of Figure 1.7(b). In Figure 1.9(b), the period-three orbit has been formed. Of course, since each point of a period-three orbit of g

is a fixed point of g^3 , the period-three orbit will appear as three intersections with the diagonal $y = x$. As you can see from the figure, the shape of the graph forces two period-three orbits to be created simultaneously. This is called a saddle-node bifurcation, or alternatively, a tangency bifurcation. The “node” is the sink, which is the set of three points at which the graph intersects the diagonal in negative slope. (Can you explain why the three negative slopes are exactly equal? Use the chain rule.) The fact that it is a sink corresponds to the fact that the negative slopes are between 1 and 0. The “saddle” is a period-three source consisting of the three upward sloping points. A vertical slice through the middle of Figure 1.7(b) shows that all initial conditions are attracted to the period-three sink. In Figure 1.9(c), the period-three sink has turned into a source. This parameter value $a = 3.86$ corresponds to the right side of Figure 1.7(b). There are many more features of Figure 1.7 that we have to leave unexplained for now. The demise of the period-three sink as an attractor coincides with a so-called period-doubling bifurcation, which creates a period-six sink, which then meets a similar fate. There are periodic windows of arbitrarily high period. We will try to unlock some of the deeper mysteries of bifurcations in Chapter 11. What happens to the bifurcation diagram if different x values are selected? (Recall that for each a , the orbit of one randomly chosen initial x is computed.) Surprisingly, nothing changes. The diagram looks the same no matter what initial

condition is picked at random between 0 and 1, since there is at most one attracting fixed or periodic orbit at each parameter value. As we shall see, however, there are many unstable, hence unseen, periodic orbits for larger a .

1.6 THE LOGISTIC MAP $G(x) = 4x(1-x)$ In the previous sections we studied maps from the logistic family $g(x) = ax(1-x)$. For $a = 2.0, 3.3$, and 3.5 , we found the existence of sinks of period 1, 2, and 4, respectively. Next, we will focus on one more case, $a = 4.0$, which is so interesting that it gets its own section. The reason that it is so interesting is that it has no sinks, which leads one to ask where orbits end up. The graph of $G(x) = 4x(1-x)$ is shown in Figure 1.10(a). Although the graph is a parabola of the type often studied in elementary precalculus courses, the map defined by G has very rich dynamical behavior. To begin with, the diagonal line $y = x$ intersects $y = G(x) = 4x(1-x)$ in the points $x = 0$ and $x = 3/4$, so there are two fixed points, both unstable. Does G have any other periodic orbits? One way to look for periodic orbits is to sketch the graph of $y = G^n(x)$. Any period-two point, for example, will be a fixed point of $G^2(x)$. Therefore we can find periodic points graphically.

The graph of $y = G^2(x)$ is shown in Figure 1.10(b). It is not hard to verify by hand the general shape of the graph. First, note that the image of $[0,1]$ under G

is $[0,1]$, so the graph stays entirely within the unit square. Second, note that $G(1/2) = 1$ and $G(1) = 0$ implies that $G^2(1/2) = 0$. Further, since $G(a_1) = 1/2$ for some a_1 between 0 and $1/2$, it follows that $G^2(a_1) = 1$. Similarly, there is another number a_2 such that $G^2(a_2) = 1$. It is clear from Figure 1.10(b) that G^2 has four fixed points, and therefore G has four points that have period either one or two. Two of these points are already known to us—they are fixed points for G . The new pair of points, p_1 and p_2 , make up a period-two orbit: that is, $G(p_1) = p_2$ and $G(p_2) = p_1$. This reasoning should have you convinced that the period-two orbit exists. The next exercise asks you to explicitly find p_1 and p_2 . EXERCISE T1.7 Find the period-two orbit of $G(x) = 4x(1-x)$. Does G have any period-three points? There is a point b_1 between 0 and a_1 for which $G(b_1) = a_1$. This implies that $G^3(b_1) = 1$. The same holds for three other points in $[0,1]$, so $y = G^3(x)$ has four relative maxima of height 1 in $[0,1]$. Since $G(1) = 0$, G^3 has roots at $x = 0, a_1, 1/2, a_2$, and 1, which separate the maxima. The graph of G^3 is shown in Figure 1.10(c). The map G^3 has eight fixed points, two of which were known to be the fixed points 0 and $3/4$ of G . The period-two points of G are not fixed points of G^3 . (Why not?) There remain six more points to account for, which must form two period-three orbits. You should be able to prove to yourself in a similar way that G^4 has $16 - 24$ fixed points, all in $[0, 1]$. With each successive iteration of G , the number of fixed points of the iterate is doubled. In general, we see that G^k has $2k$ fixed points, all in $[0, 1]$. Of course, for $k = 1$, G has fewer than $2k$ points of period- k . (Remember that the definition of period- k for the point p is that k is the smallest positive integer for which $f^k(p) = p$.) For example, $x = 0$ is a period one point and therefore not a period- k point for $k = 1$, although it is one of the $2k$ fixed points of G^k . EXERCISE T1.8 Let $G(x) = 4x(1-x)$. Prove that for each positive integer k , there is an orbit of period- k .

The number of orbits of the map for each period can be tabulated in the map's periodic table. For the logistic map it begins as shown in Table 1.3. The first column is the period k , and the second column is the number of fixed points of f^k , which is $2k$, as seen in Figure 1.10. The third column keeps track of fixed points of G^k which correspond to orbits of lower period than k . When these are subtracted away from the entry in the second column, the result is the number of period- k points, which is divided by k to get the number of period- k orbits. EXERCISE T1.9 Let $G(x) = 4x(1-x)$. (a) Decide whether the fixed points and period-two points of G are sinks. (b) Continue the periodic table for G begun in Table 1.3. In particular, how many periodic orbits of (minimum) period k does G have, for each $k \geq 10$? Is this what we mean by "chaos"? Not exactly. The existence of infinitely many periodic orbits does not in itself imply the kind of unpredictability usually associated with chaotic maps, although it does hint at the rich structure present. Chaos is identified with nonperiodicity and sensitive dependence on initial conditions, which we explore in the next section.

1.7 SENSITIVE DEPENDENCE ON INITIAL CONDITION
EXAMPLE 1.9 Consider the map $f(x) = 3x \pmod{1}$ on the unit interval. The notation $y \pmod{1}$ stands for the number $y - n$, where n is the unique integer that makes $0 \leq y - n < 1$. For example, $14.92 \pmod{1} = .92$ and $14.92 \pmod{1} = .08$. For a positive number y , this is the fractional part of y . See Figure 1.11(a) for

a graph of the map. Because of the breaks at $x = 1/3, 2/3$, this function is not continuous. This map is not continuous, however the important property that we are interested in is not caused by the discontinuity. It may be more natural to view f as a map on the circle of circumference one. Glue together the ends of the unit interval to form a circle, as in Figure 1.11(b). If we consider $f(x)$ as a map from this circle to itself, it is a continuous map. In Figure 1.11(b), we show the image of the subinterval $[0, 1/2]$ on the circle. Whether we think of f as a discontinuous map on the unit interval or as a continuous map on the circle makes no difference for the questions we will try to answer below. We call a point x eventually periodic with period p for the map f if for some positive integer N , $f^n(x) = f^{n+p}(x)$ for all $n \geq N$, and if p is the smallest

such positive integer. This says exactly that the orbit of x eventually maps directly onto a periodic orbit. For example, $x = 1/3$ is an eventually periodic point, since it maps under f to the period one orbit 0. EXERCISE T1.10 Show that a point x is eventually periodic for Example 1.9 if and only if x is a rational number. EXERCISE T1.11 Construct the periodic table for f in Example 1.9 (follow the form given by Table 1.3). The $3x \bmod 1$ map demonstrates the main characteristic of chaos: sensitive dependence on initial conditions. This refers to the property that pairs of points, which begin as close together as desired, will eventually move apart. Table 1.4 shows the beginning of two separate orbits whose initial conditions differ by .0001. In fact, no matter how close they begin, the difference between two nearby orbits is—as measured on the circle—magnified by a factor of 3 on each iteration. This idea is important enough to be assigned a formal definition.

Definition 1.10 Let f be a map on S . A point x_0 has sensitive dependence on initial conditions if there is a nonzero distance d such that some points arbitrarily near x_0 are eventually mapped at least d units from the corresponding image of x_0 . More precisely, there exists $d > 0$ such that any neighborhood N of x_0 contains a point x such that $|f^k(x) - f^k(x_0)| \geq d$ for some nonnegative integer k . Sometimes we will call such a point x_0 a sensitive point. Ordinarily, the closer x is to x_0 , the larger k will need to be. The point x will be sensitive if it has neighbors as close as desired that eventually move away the prescribed distance d for some sufficiently large k . EXERCISE T1.12 Consider the $3x \bmod 1$ map of the unit interval $[0, 1]$. Define the distance between a pair of points x, y to be either $|x - y|$ or $1 - |x - y|$, whichever is smaller. (We are measuring with the “circle metric”, in the sense of Figure 1.11, corresponding to the distance between two points on the circle.) (a) Show that the distance between any pair of points that lie within $1/6$ of one another is tripled by the map. (b) Find a pair of points whose distance is not tripled by the map. (c) Show that to prove sensitive dependence for any point, d can be taken to be any positive number less than $1/2$ in Definition 1.10, and that k can be chosen to be the smallest integer greater than $\ln(d - |x - x_0|) / \ln 3$. EXERCISE T1.13 Prove that for any map f , a source has sensitive dependence on initial conditions. 1.8 I T I N E R A R I E S The fact that the logistic map $G(x) = 4x(1 - x)$ has periodic orbits of every period is one indication of its complicated dynamics. An even more important reflection of this complexity is sensitive dependence on initial

conditions, which is the hallmark of chaos. In this section we will show for the logistic map G_4 that for any initial point in the unit interval and any preset distance $\epsilon > 0$, no matter how small, there is a second point within ϵ units of the first so that their two orbits will map at least $d - 1/4$ units apart after a sufficient number of iterations. Since $1/4$

unit is 25 conditions which began very close to one another are eventually moved by the map so they are no longer close, by any reasonable definition of “close”. In order to investigate sensitive dependence, we introduce the concept of the itinerary of an orbit. This is a bookkeeping device that allows much of the information of an orbit to be coded in terms of discrete symbols. For the logistic map, assign the symbol L to the left subinterval $[0, 1/2]$, and R to the right subinterval $(1/2, 1]$. Given an initial value x_0 , we construct its itinerary by listing the subintervals, L or R, that contain x_0 and all future iterates. For example, the initial condition $x_0 = 1/3$ begets the orbit $1/3, 8/9, 32/81, \dots$, whose itinerary begins LRL $\dots$. For the initial condition $x_0 = 1/4$, the orbit is $1/4, 3/4, 3/4, \dots$,

which terminates in the fixed point $x = 3/4$. The itinerary for this orbit is LRR $\dots$, which we abbreviate by LR; the overbar indicates that the R repeats indefinitely. Notice that there is a special orbit, or group of orbits, for which the itinerary is not uniquely defined. That is because the intervals L and R overlap at $x = 1/2$. In particular, consider the initial condition $x_0 = 1/2$. The corresponding orbit is $1/2, 1, 0, 0, \dots$, which can be assigned itinerary RRL or LRL. This particular orbit (and some others like it) are assigned two different names under this naming system. Except for the case of orbits which land precisely on $x = 1/2$ at some point of the orbit (and therefore end up mapping onto the fixed point 0), the itinerary is uniquely defined. Once we are given this way of assigning an itinerary to each orbit, we can map out, on the unit interval, the locations of points that have certain itineraries. Of course, an itinerary is in general an infinite sequence, but we could ask: what is the set of points whose itinerary begins with, say, LR? These points share the property of beginning in the L subinterval and being mapped to the R subinterval by one iterate of the map. This set, which we could call the LR set, is shown in Figure 1.12, along with a few other similar sets. We would like to identify the sets of all initial points whose itineraries begin with a specified sequence of symbols. For example, the set of initial conditions whose itinerary begins with LR forms a subinterval of the unit interval. The subintervals in Figure 1.12 give information about the future behavior of the initial conditions lying in them. Another example is the subinterval marked LRL, which consists of orbits that start out in the interval L $[0, 1/2]$, whose first iterate lands in R $(1/2, 1]$, and whose second iterate lands in $[0, 1/2]$. For example, $x = 1/3$ lies in LRL. Likewise, $x = 1/4$ lies in LRR because its first and second iterate are in R. EXERCISE T1.14 (a) Find a point that lies in the subinterval LLR. (You are asked for a specific number.) (b) For each subinterval corresponding to a sequence of length 3, find a point in the subinterval. You may see some patterns in the arrangement of the subintervals of Figure 1.12. It turns out that the rule for dividing an interval, say LR, into its two subintervals is the following: Count the number of R's in the sequence

(one in this case). If odd, the interval is divided into LRR, LRL in that order. If even, the L subinterval precedes the R subinterval. With this information, the reader can continue Figure 1.12 schematically to finer and finer levels.

EXERCISE T1.15 Continue the schematic diagram in Figure 1.12 to subintervals corresponding to length 4 sequences. EXERCISE T1.16 Let x_0 be a point in the subinterval RLLRRRLRLR. (a) Is x_0 less than, equal to, or greater than $1/2$? (b) Same question for $f^6(x_0)$. A graphical way of specifying the possible itineraries for the logistic map is shown in Figure 1.13. We call this the transition graph for the subintervals L and R. An arrow from L to R, for example, means that the image $f(L)$ contains the interval R. For every path through this graph with directed edges (arrows), there exists an orbit with an itinerary satisfying the sequence of symbols determined by the path. It is clear from Figure 1.12 that the image of each of the intervals L and R contains both L and R, so the transition graph for the logistic map is fully connected (every symbol is connected to every other symbol by an arrow). Since the graph is fully connected, all possible sequences of L and R are possible. The concept of itineraries makes it easy to explain what we mean by sensitive dependence on initial conditions. In specifying the first k symbols of the itinerary, we have 2^k choices. If k is large, then most of the 2^k subintervals are forced to be rather small, since the sum of their lengths is 1. It is a fact (that we will prove in Chapter 3) that each of the 2^k subintervals is shorter than 2^{-k} in length. Consider any one of these small subintervals for a large value of k , corresponding to some sequence of symbols $S_1 \dots S_k$, where each S_i is either R or L. This subinterval in turn contains subintervals corresponding to the sequences $S_1 \dots S_k LL$, $S_1 \dots S_k LR$, $S_1 \dots S_k RR$ and $S_1 \dots S_k RL$. If we choose one point from each, we have four initial conditions that lie within 2^{-k} (since they all lie

in $S_1 \dots S_k$), but which map k iterates later to subintervals LL, LR, RR, and RL, respectively. (If this step isn't clear, it may help to recall Exercise T1.16.) In Figure 1.12, the width of the LR and RR subintervals are greater than $1/4$, so that the LR and RL subintervals, for example, lie over $1/4$ unit apart. It is now possible to see why every point in $[0, 1]$ has sensitive dependence on initial conditions under the logistic map G . To find a neighbor close to x_0 that eventually separates by a distance of at least $d = 1/4$, identify which subinterval of level $k = 2$ that x_0 belongs to, say $S_1 \dots S_k LR$. Then it is always possible to identify a subinterval within 2^{-k} which maps $1/4$ unit away after k iterates, such as $S_1 \dots S_k RL$. Therefore every point exhibits sensitive dependence with neighbors that are arbitrarily close. We illustrate for $k = 1000$: There is a pair of initial conditions within $2^{-1000} \approx 10^{-300}$ that eventually are mapped at least $1/4$ unit apart. This is an expansion of 1000 factors of 2 in 1000 iterates, for an average multiplicative expansion rate of approximately 2. In Chapter 3 we will introduce the term "Lyapunov number", which will quantify the average multiplicative separation rate of a map, which is in this case $2^{1000/1000} = 2$ per iterate. The fact that this number is greater than 1 will mean that repeated expansion is occurring. The impact of sensitive dependence is that changes in initial measurements or errors in calculation along the orbit can result in quite different outcomes. The consequences of this behavior were not fully appreciated

until the advent of computers and the computer simulation of dynamical models.

C O M P U T E R E X P E R I M E N T 1 . 4 Use a computer program to illustrate sensitive dependence for the logistic map $G(x) = 4x(1-x)$. Start with two different initial conditions that are very close together, and iterate G on each. The two orbits should stay near one another for a while, and then separate for good. By collecting statistics on your experiments, try to quantify how many iterations are required for the points to move apart, say 1/2 unit, when the initial separation is .01, .001, etc. Does the location of the initial pair matter?

Period Three Implies Chaos IN CHAPTER 1 we have studied periodic orbits for continuous maps and the idea of sensitive dependence on initial conditions. In Challenge 1 you will prove the fact that the existence of a period-three orbit alone implies the existence of a large set of sensitive points. The set is infinite (in fact, uncountably infinite, a concept we will study in more detail later in the book). This surprising fact was discovered by T.Y. Li and J.A. Yorke (Li and Yorke, 1975). A chaotic orbit is a bounded, non-periodic orbit that displays sensitive dependence. When we give a precise definition of chaos, we will find that the discussion is simplified if we require a stronger definition of sensitivity, namely that chaotic orbits separate exponentially fast from their neighbors as the map is iterated. A much simpler fact about continuous maps is that the existence of a period-three orbit implies that the map has periodic orbits of all periods (all integers). See Exercise T3.10 of Chapter 3. This fact doesn't say anything directly about sensitive dependence, although it guarantees that the map has rather complicated dynamical behavior. We show a particular map f in Figure 1.14 that has a period-three orbit, denoted A, B, C . That is, $f(A) = B$, $f(B) = C$, and $f(C) = A$. We will discover that there are infinitely many points between A and C that exhibit sensitive dependence on initial conditions. To simplify the argument, we will use an assumption that is explicitly drawn into Figure 1.14: the map $f(x)$ is unimodal, which means that it has only one critical point. (A critical point for a function $f(x)$ is a point for which $f'(x) = 0$ or where the derivative does not exist.) This assumption, that $f(x)$ has a single maximum, is not necessary to prove sensitive dependence—in fact sensitive dependence holds for any continuous map with a period-three orbit. The final step of Challenge 1 asks you to extend the reasoning to this general case. The existence of the period-three orbit in Figure 1.14 and the continuous nature of f together guarantee that the image of the interval $[A, B]$ covers $[B, C]$; that is, that $f[A, B] \supset [B, C]$. Furthermore, $f[B, C] \supset [A, C]$. We will try to repeat our analysis of itineraries, which was successful for the logistic map, for this new map. Let the symbol L represent the interval $[A, B]$, and R represent $[B, C]$.

Unlike the logistic map example, notice that in the itinerary of an orbit, L must be followed by R , although R can be followed by either L or R . Some of the itinerary subintervals are shown schematically in Figure 1.15. A second difference from the logistic map example is that there may be gaps in the interval, as shown in Figure 1.15. For example, points just to the left of B are mapped to the right of C , and therefore out of the interval $[A, C]$; we do not include these points in our analysis. (A more sophisticated analysis might include these points, but we can demonstrate sensitive dependence without considering

these orbits.) To simplify our analysis, we will not assign an itinerary to points that map outside $[A, C]$. The corresponding transition graph is shown in Figure 1.16. The transition graph tells us that every finite sequence of the symbols L and R corresponds to a subinterval of initial conditions x , as long as there are never two consecutive L's in the sequence. (This follows from the fact that the left-hand interval $[A, B]$ does not map over itself.) The proof that period three implies chaos is given below in outline form. In each part, you are expected to fill in a reason or argument. Step 1 Let d denote the length of the subinterval RR . Denote by $J = S_1 \dots S_k R$ any subinterval that ends in R . (Each S_i denotes either R or L .) Show

that J contains a pair of points, one in each of the subintervals $S_1 \dots S_k RRL$ and $S_1 \dots S_k RLR$, that eventually map at least d units apart. Step 2 Show that inside J there are 3 subintervals of type $S_1 \dots S_k R S_{k+2} S_{k+3} R$. Explain why at least 2 of them must have length that is less than half the length of J .

Step 3 Combine the two previous steps. Show that a subinterval J of form $S_1 \dots S_k R$ must contain a subinterval J_1 of form $S_1 \dots S_k R S_{k+2} S_{k+3} RLR$ with the following property. Each point x of J_1 has a neighbor y within $\text{length}(J)/2$ whose pairwise distance upon further iteration eventually exceeds d . Step 4 Let $h = CA$ be the length of the original interval. Show that for each positive integer k there are 2^k disjoint subintervals (denoted by sequences of $5k + 1$ symbols) of length less than $2kh$, each of which contain a point that has sensitive dependence on initial conditions. Therefore, there are infinitely many sensitive points. Step 5 Quantify the number of sensitive points you have located in the following way. Show that there is a one-to-one correspondence between the sensitive points found in Step 4 and binary numbers between 0 and 1 (infinite sequences of form $.a_1 a_2 a_3 \dots$, where each a_i is a 0 or 1). This means that the set of sensitive points is uncountable, a concept we will meet in Chapter 4. Step 6 Our argument is based on Figure 1.14, where $f(A) = B$, $f(B) = C$, $f(C) = A$, and where $A \subset B \subset C$. How many other distinct "cases" need to be considered? Does a similar argument work for these cases? What changes are necessary? Step 7 Explain how to modify the arguments above to work for the case where f is any continuous map with a period-three orbit. (Begin by identifying one-piece subintervals of $[A, B]$ and $[B, C]$ that are mapped onto $[A, B]$ and $[B, C]$.) Postscript. The subintervals described in the previous argument, although many in number, may comprise a small proportion of all points in the interval $[A, C]$. For example, the logistic map $g(x) = 3.83x(1 - x)$ has a period-three sink. Since there is a period-three orbit (its stability does not matter), we now know that there are many points that exhibit sensitive dependence with respect to their neighbors. On the other hand, the orbits of most points in the unit interval converge to one or another point in this periodic attractor under iteration by g^3 . These points do not exhibit sensitive dependence. For example, points that lie a small distance from one of the points p of the period-three sink will be attracted toward p , as we found in Theorem 1.5. The distances between points that start out near p decrease by a factor of approximately $|(g^3)'(p)|$ with each three iterates. These nearby points do not separate under iteration. There are, however, infinitely many points whose orbits do not converge to the period-three sink. It is these points that exhibit

sensitive dependence

EXERCISES 1.1. Let $l(x) = ax + b$, where a and b are constants. For which values of a and b does l have an attracting fixed point? A repelling fixed point?

1.2. (a) Let $f(x) = x - x^2$. Show that $x = 0$ is a fixed point of f , and describe the dynamical behavior of points near 0. (b) Let $g(x) = \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Show that $x = 0$ is a fixed point of g , and describe the dynamical behavior of points near 0. (c) Give an example of a function h for which $h(0) = 1$ and $x = 0$ is an attracting fixed point. (d) Give an example of a function h for which $h(0) = 1$ and $x = 0$ is a repelling fixed point.

1.3. Let $f(x) = x^3 - x$. Find all fixed points of f and decide whether they are sinks or sources. You will have to work without Theorem 1.5, which does not apply.

1.4. Let $x_1, \dots, x_8$ be the eight fixed points of $G^3(x)$, where $G(x) = 4x(1 - x)$, as in 1.10(c). Clearly, $x_1 = 0$. (a) For which i is $x_i = 3/4$? (b) Group the remaining six points into two orbits of three points each. It may help to consult Figure 1.10(c). The most elegant solution (that we know of) uses the chain rule.

1.5. Is the period-two orbit of the map $f(x) = 2x^2 - 5x$ on a sink, a source, or neither? See Exercise T1.5.

1.6. Define the map $f(x) = 2x \pmod{1}$ on the unit interval $[0, 1]$. Let L denote the subinterval $[0, 1/2]$ and R the subinterval $(1/2, 1]$. (a) Draw a chart of the itineraries of f as in Figure 1.12. (b) Draw the transition graph for f . (c) Establish sensitive dependence for orbits under this map. Show that each point has neighbors arbitrarily near that eventually map at least $1/2$ unit apart.

1.7. Define the tent map on the unit interval $[0, 1]$ by $T(x) = 2x$ if $0 \leq x \leq 1/2$ and $T(x) = 2(1 - x)$ if $1/2 \leq x \leq 1$. (a) Divide the unit interval into two appropriate subintervals and repeat parts (a)–(c) of Exercise 1.6 for this map.

(b) Complete a periodic table for f , similar to the one in Table 1.3, for periods less than or equal to 10. In what ways, if any, does it differ from the periodic table for the logistic map G ?

1.8. Let $f(x) = 4x(1 - x)$. Prove that there are points in $I = [0, 1]$ that are not fixed points, periodic points, or eventually periodic points of f .

1.9. Define $x_{n+1} = (x_n - 2)/(x_n - 1)$. (a) Find $L = \lim_{n \rightarrow \infty} x_n$ for $x_0 \neq 0$. (b) Describe the set of all negative x_0 for which the limit $\lim_{n \rightarrow \infty} x_n$ either exists and is not equal to the L in part (a) or does not exist (for example, $x_0 = 1$).

1.10. For the map $g(x) = 3.05x(1 - x)$, find the stability of all fixed points and period-two points.

1.11. Let f be a one-to-one smooth map of the real line to itself. One-to-one means that if $f(x_1) = f(x_2)$, then $x_1 = x_2$. A function f is called increasing if $x_1 < x_2$ implies $f(x_1) < f(x_2)$, and decreasing if $x_1 < x_2$ implies $f(x_1) > f(x_2)$. (a) Show that f is increasing for all x or f is decreasing for all x . (b) Show that every orbit $x_0, x_1, x_2, \dots$ of f^2 satisfies either $x_0 \leq x_1 \leq x_2 \leq \dots$ or $x_0 \geq x_1 \geq x_2 \geq \dots$. (c) Show that every orbit of f^2 either diverges to $-\infty$ or $+\infty$ or converges to a fixed point of f^2 . (d) What does this imply about convergence of the orbits of f ?

1.12. The map $g(x) = 2x(1 - x)$ has negative values for large x . Population biologists sometimes prefer maps that are positive for positive x . (a) Find out for what value of a the map $h(x) = axex$ has a superstable fixed point x_0 , which means that $h(x_0) = x_0$ and $h'(x_0) = 0$. (b) Investigate the orbit starting at $x_0 = 0.1$ for this value of a using a calculator. How does the behavior of this orbit differ if a is increased by 50? (c) What is the range of a for which $h(x)$ has a positive sink?

1.13. Let $f : [0, \infty) \rightarrow [0, \infty)$ be a smooth function, $f(0) = 0$, and let $p > 0$ be

a fixed point such that $f'(p) < 0$. Assume further that $f(x)$ is decreasing. Show that all positive x_0 converge to p under f . 1.14. Let $f(x) = x^2 - x$. Find all fixed points of f . Where do nonfixed points go under iteration by f ? 1.15. Prove the following explicit formula for any orbit $x_0, x_1, x_2, \dots$ of the logistic map $G(x) = 4x(1-x)$: $x_n = \frac{1}{2} - \frac{1}{2} \cos(2^n \arccos(1 - 2x_0))$. Caution: Not all explicit formulas are useful.

1.16. Let f be a map defined on the real line, and assume p is a fixed point. Let $\epsilon > 0$ be a given number. Find a condition that guarantees that every initial point x in the interval $(p - \epsilon, p + \epsilon)$ satisfies $f^n(x) \rightarrow p$ as $n \rightarrow \infty$. 1.17. Let $f(x) = 4x(1-x)$. Prove that $\bigcup_{k=0}^{\infty} I_k$, the interval of initial values x in $[0, 1]$ such that $0 < f^i(x) < \frac{1}{2}$ for $0 \leq i < k$, has length $[1 - \cos(2^{-k})]^{-1}$.